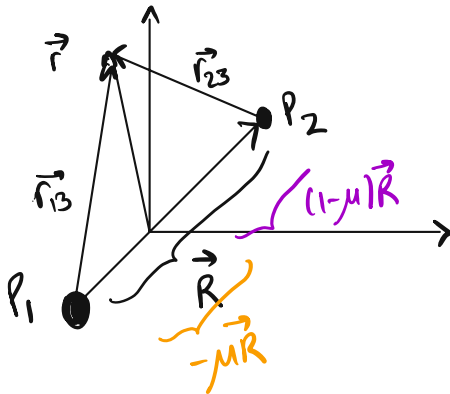


(R3BP)I. INTRODUCTION

$$\ddot{\vec{r}} = -G(M_1 + M_2) \left[ \frac{(1-\mu)\vec{r}_{13}}{|\vec{r}_{13}|^3} + \mu \frac{\vec{r}_{23}}{|\vec{r}_{23}|^3} \right]$$



$$\begin{aligned}\vec{r}_{13} &= \vec{r} + \mu \vec{R} \\ \vec{r}_{23} &= \vec{r} - (1-\mu)\vec{R}\end{aligned}$$

$$\begin{aligned}\mu &= \frac{m_2}{m_1 + m_2} \\ 1-\mu &= \frac{m_1}{m_1 + m_2}\end{aligned}$$

- Let  $\vec{R}$  be in a circular orbit:

$$|\dot{\vec{R}}| = R = \text{const}$$

$$|\dot{\vec{R}}| = \sqrt{\frac{G(M_1 + M_2)}{R}}$$

$$n = \sqrt{G(M_1 + M_2)/R^3}$$

I.I. Dimensional EoM in an Inertial Frame

$$\ddot{\vec{r}} = -G(M_1 + M_2) \left[ \frac{(1-\mu)(\vec{r} + \mu\vec{R})}{|\vec{r} + \mu\vec{R}|^3} + \mu \frac{\vec{r} - (1-\mu)\vec{R}}{|\vec{r} - (1-\mu)\vec{R}|^3} \right] = \frac{\partial U}{\partial \vec{r}}$$

$$U = G(M_1 + M_2) \left[ \frac{(1-\mu)}{|\vec{r} + \mu\vec{R}|} + \frac{\mu}{|\vec{r} - (1-\mu)\vec{R}|} \right]$$

I.II. Non-dimensional EoM in an Inertial Frame

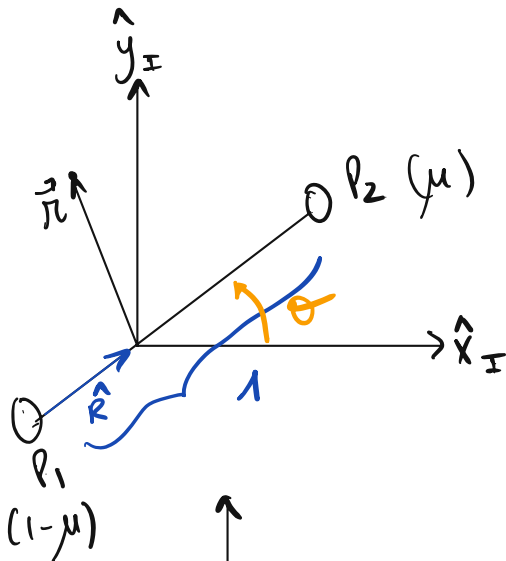
- Non-dimensionalize: (i) Length scale  $R$ ,  $\vec{r} = \vec{r}/R$ ,  $\hat{\vec{R}} = \vec{R}/R$  ↖ non-dim position
- (ii) Time scale Mean anomaly of  $P_1, P_2$   $M = nt$  ↘ non-dim time  
 $\frac{1}{n}$ ,  $\theta = \frac{t}{(1/n)} = nt$

- $R\ddot{\vec{r}} = -\frac{G(M_1+M_2)R}{R^3} \left[ \frac{(1-\mu)(\vec{r} + \mu\hat{R})}{|\vec{r} + \mu\hat{R}|^3} + \frac{\mu(\vec{r} - (1-\mu)\hat{R})}{|\vec{r} - (1-\mu)\hat{R}|^3} \right]$
- $\frac{d\vec{r}}{dt} = \frac{d\vec{r}}{d\theta} \frac{d\theta}{dt} = n \frac{d\vec{r}}{d\theta}$  ;  $\theta = nt$ ,  $d\theta = ndt$ ,  $\frac{d\theta}{dt} = n$
- $\frac{d^2\vec{r}}{dt^2} = \frac{d^2\vec{r}}{d\theta^2} n^2$  ;  $n^2 = \frac{G(M_1+M_2)}{R^3} \Rightarrow \text{cancels out on rhs}$
- $\frac{d\vec{r}}{dt} = \dot{\vec{r}}$  ;  $\frac{d\vec{r}}{d\theta} = \vec{r}'$  ;  $\frac{d^2\vec{r}}{d\theta^2} = \vec{r}''$

$$\vec{r}'' = - \left[ \frac{(1-\mu)(\vec{r} + \mu\hat{R})}{|\vec{r} + \mu\hat{R}|^3} + \frac{\mu(\vec{r} - (1-\mu)\hat{R})}{|\vec{r} - (1-\mu)\hat{R}|^3} \right] = \frac{\partial U_{\text{ND}}}{\partial \vec{r}}$$

$$U_{\text{ND}} = \frac{1-\mu}{|\vec{r} + \mu\hat{R}|} + \frac{\mu}{|\vec{r} - (1-\mu)\hat{R}|}$$

### I.III. Non-Dimensional EoM in a Rotating Frame



- $\hat{R}$  is a time-varying vector
- Choose coordinate frame such that  $\hat{z}_I = \hat{n}$ , then  $\hat{x}_I, \hat{y}_I$  are in the  $P_1, P_2$  orbit plane and:

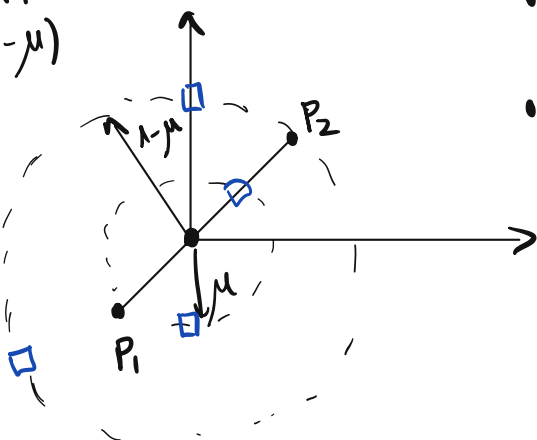
$$\hat{R} = \cos(\theta)\hat{x}_I + \sin(\theta)\hat{y}_I = \cos(nt)\hat{x}_I + \sin(nt)\hat{y}_I$$

- $U_{\text{ND}}(\vec{r}, \theta)$  ;  $U_{\text{DIN}}(\vec{r}, t)$

$$E = \frac{1}{2} \vec{r}' \cdot \vec{r}' - U(\vec{r}, \theta)$$

$\Rightarrow E$  is not conserved.

; and



- Recall HW #1, if we can find a potential  $V(\vec{r})$  that's not a function of time, then we will conserve "modified energy"  $E = \frac{1}{2} \dot{r}^2 - V$ .
- Choose a uniformly rotating frame that rotates with  $P_1 - P_2$  vector  $\hat{R}$ .
- It rotates at  $\vec{\Omega} = \hat{z}$   $\Rightarrow$

$$T_{I \rightarrow BF}(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\hat{R} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix} \Rightarrow T_{I \rightarrow BF} \cdot \hat{R} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \hat{R} = \hat{x}_R$$

$$\Rightarrow V(\vec{r}) = \frac{1-\mu}{|\vec{r} + \mu \hat{x}_R|} + \frac{\mu}{|\vec{r} - (1-\mu) \hat{x}_R|}$$

Constant in the  
rot. frame (for fixed  $\vec{r}$ )

- We must rewrite the EoM

Transport Theorem

$$\vec{r}'_I = (\vec{r}')_R + \vec{\Omega} \times \vec{r}_R$$

$$\vec{r}''_I = \vec{r}''_R + 2\vec{\Omega} \times \vec{r}'_R + \vec{\Omega} \times \vec{\Omega} \times \vec{r} + \vec{\Omega}' \times \vec{r}$$

- Algebraic manipulations:

$$\vec{r}''_R + \underbrace{2\vec{\Omega} \times \vec{r}'_R}_{\text{Coriolis}} + \underbrace{\vec{\Omega} \times \vec{\Omega} \times \vec{r}}_{\text{centrifugal}} = \underbrace{\frac{\partial V_{ND,R}}{\partial \vec{r}}}_{\text{gravity}}$$

$$\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; \vec{\Omega} \times \vec{r}' = \begin{bmatrix} -y' \\ -x' \\ 0 \end{bmatrix} ; \vec{\Omega} \times \vec{\Omega} \times \vec{r} = \begin{bmatrix} -x \\ -y \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} + 2 \begin{bmatrix} -y' \\ x' \\ 0 \end{bmatrix} - \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} \partial U / \partial x \\ \partial U / \partial y \\ \partial U / \partial z \end{bmatrix}$$

$\underbrace{\quad}_{\frac{\partial}{\partial(x,y,z)} \left( -\frac{1}{2}(x^2+y^2) \right)}$

$$x'' - 2y' = x + \frac{\partial U}{\partial x} = \frac{\partial}{\partial x} \left[ \underbrace{-\frac{1}{2}(x^2+y^2)}_V + U \right] = \frac{\partial}{\partial x} V$$

$$y'' + 2x' = y + \frac{\partial U}{\partial y} = \frac{\partial}{\partial y} V$$

$$z'' = \frac{\partial V}{\partial z}$$

$$V = \frac{1}{2}(x^2+y^2) + U \quad U = \frac{(1-\mu)}{[(x+\mu)^2+y^2+z^2]^{1/2}} + \frac{\mu}{[(x-(1-\mu))^2+y^2+z^2]^{1/2}}$$

"standard form" of the CRBP in a rotating frame,  
non-dimensional version.

### I. IV. Dimensional EoM in a Rotating Frame

$$\ddot{x} - 2n\dot{y} = \frac{\partial V}{\partial x}$$

$$V = \frac{1}{2}n^2(x^2+y^2) + U$$

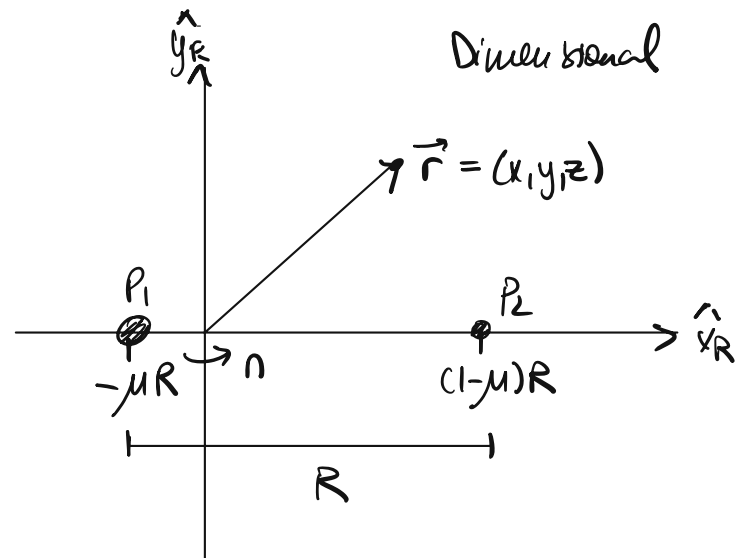
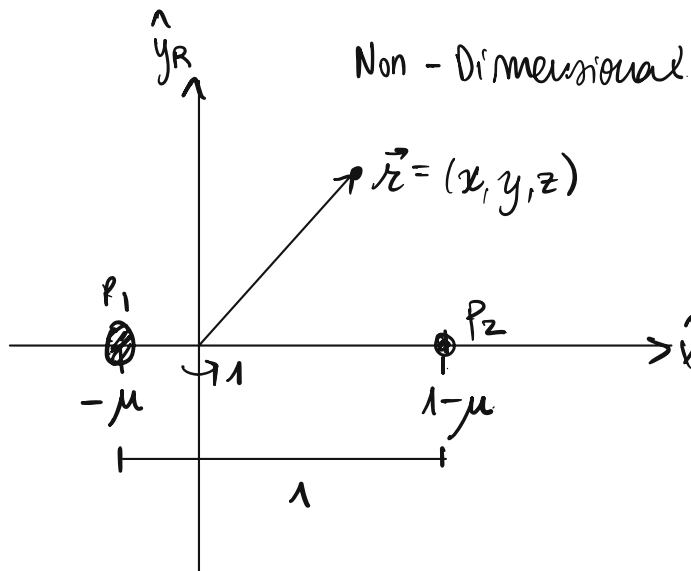
$$\ddot{y} + 2n\dot{x} = \frac{\partial V}{\partial y}$$

$$U = G(M_1+M_2) \left[ \frac{(1-\mu)}{[(x+\mu R)^2+y^2+z^2]^{1/2}} + \frac{\mu}{[(x-(1-\mu)R)^2+y^2+z^2]^{1/2}} \right]$$

$$\ddot{z} = \frac{\partial V}{\partial z}$$

$$n = \sqrt{\frac{G(M_1+M_2)}{R^3}}$$





## II. JACOBI INTEGRAL

### II.I Jacobi integral in the rotating frame

- $V = \frac{1}{2} (x^2 + y^2) + V(x, y, z)$  is time invariant if we fix  $\vec{r}$

Claim "Energy Integral" exists:  $E = \frac{1}{2} \vec{r}' \cdot \vec{r}' - V(x, y, z)$

Proof

$$\textcircled{1} x'' - 2y' - V_x = 0 ; \textcircled{2} y'' + 2x' - V_y = 0 ; \textcircled{3} z'' - V_z = 0$$

$$x' \cdot \textcircled{1} + y' \cdot \textcircled{2} + z' \cdot \textcircled{3} = 0$$

$$x'x'' - \cancel{2x'y'} - V_x x' + y'y'' + \cancel{2y'x'} - V_y y' + z'z'' - V_z z' = 0$$

$$\frac{1}{2} \frac{d}{d\sigma} (x'^2 + y'^2 + z'^2) - \frac{d}{d\sigma} V(x, y, z) = 0$$

$$\frac{d}{d\sigma} \left( \underbrace{\frac{1}{2} \vec{r}' \cdot \vec{r}' - V(x, y, z)}_{\text{Jacobi Constant}} \right) = 0$$

□

Definition

$$J = \frac{1}{2} \dot{\vec{r}}' \cdot \dot{\vec{r}}' - V(\vec{r}) \equiv \text{Jacobi constant}$$

$$C_J = -2J$$

$$\underbrace{\frac{1}{2} \dot{\vec{r}}' \cdot \dot{\vec{r}}'}_{\text{KE rel to rot frame}} - \underbrace{\frac{1}{2} (x^2 + y^2)}_{\text{PE due to rot frame}} - \underbrace{U(\vec{r})}_{\text{grav pot.}}$$

Note Poincaré proved that there are no other integrals of motion in a simple form, defined everywhere

Note  $J$  does not exist for the  $E, P, H$ , restricted 3BPs.

Formulas:

Dimensionless

(Rotating frame)

$$\begin{aligned} J &= \frac{1}{2} \dot{\vec{r}}' \cdot \dot{\vec{r}}' - V(\vec{r}) \\ V &= \frac{1}{2} (x^2 + y^2) + U(\vec{r}) \\ U &= \frac{1-\mu}{|\vec{r} + \mu \hat{x}|} + \frac{\mu}{|\vec{r} - (1-\mu) \hat{x}|} \end{aligned}$$

Dimensional

(Rotating frame)

$$\begin{aligned} J &= \frac{1}{2} \dot{\vec{r}} \cdot \dot{\vec{r}} - V(\vec{r}) \\ V &= \frac{1}{2} n^2 (x^2 + y^2) + U(\vec{r}) \\ U &= G(M_1 + M_2) \left[ \frac{1-\mu}{|\vec{r} + \mu R \hat{x}|} + \frac{\mu}{|\vec{r} - (1-\mu) R \hat{x}|} \right] \end{aligned}$$

## II.II. Jacobi integral in the Inertial Frame

- There should be a form of  $J$  in the inertial frame!
- Take the Jacobi Integral and replace  $\dot{\vec{r}}_R$  with  $\dot{\vec{r}}_I \equiv \dot{\vec{v}}_I$

$$\dot{\vec{r}}_R = \dot{\vec{v}}_I - n \hat{z} \times \vec{r}$$

$$J = \frac{1}{2} \dot{\vec{r}} \cdot \dot{\vec{r}} - V$$

$$\begin{aligned} \Rightarrow \frac{1}{2} \dot{\vec{r}} \cdot \dot{\vec{r}} &= \frac{1}{2} (\dot{\vec{v}} - n \hat{z} \times \vec{r}) \cdot (\dot{\vec{v}} - n \hat{z} \times \vec{r}) = \frac{1}{2} v_I^2 + \frac{1}{2} n^2 (\hat{z} \times \vec{r}) \cdot (\hat{z} \cdot \vec{r}) - n \dot{\vec{v}}_I \cdot (\hat{z} \times \vec{r}) \\ &= \frac{1}{2} v^2 - n \hat{z} \cdot (\vec{r} \times \dot{\vec{v}}) + \frac{1}{2} n^2 (x^2 + y^2) - \frac{1}{2} \dot{\vec{r}} \cdot \dot{\vec{r}} \end{aligned}$$

$$V = \frac{1}{2} n^2 (x^2 + y^2) + U$$

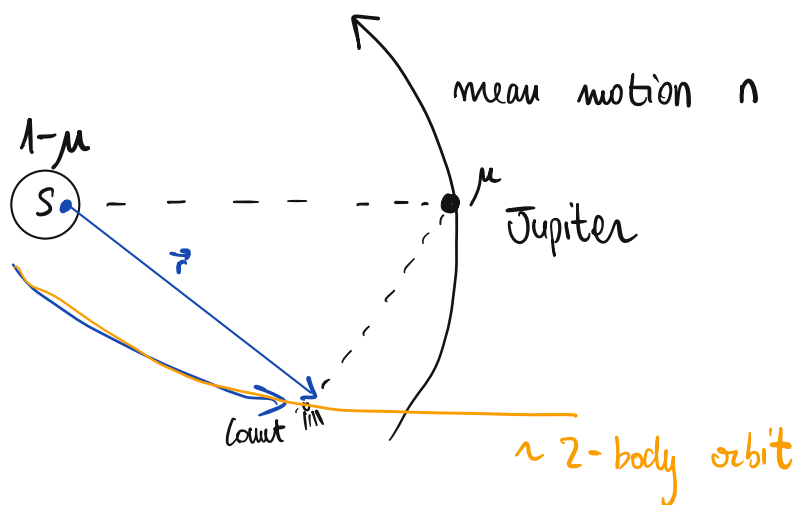
$$\Rightarrow J = \underbrace{\frac{1}{2} v^2 - U(\vec{r}, t)}_{\text{Specific Energy}} - \underbrace{n \hat{z} \cdot (\vec{r} \times \vec{v})}_{\text{Angular Mom}} = \text{Constant}$$

Inertial Frame

### III. TISSERAND CRITERION

- $J$  can be rewritten in terms of osculating orbital elements, if we make an assumption or two.

$$J = \frac{1}{2} v^2 - G(M_1 + M_2) \left[ \frac{1-\mu}{|\vec{r} + \mu \vec{R}|} + \frac{\mu}{|\vec{r} - (1-\mu)\vec{R}|} \right] - n \hat{z} \cdot (\vec{r} \times \vec{v})$$

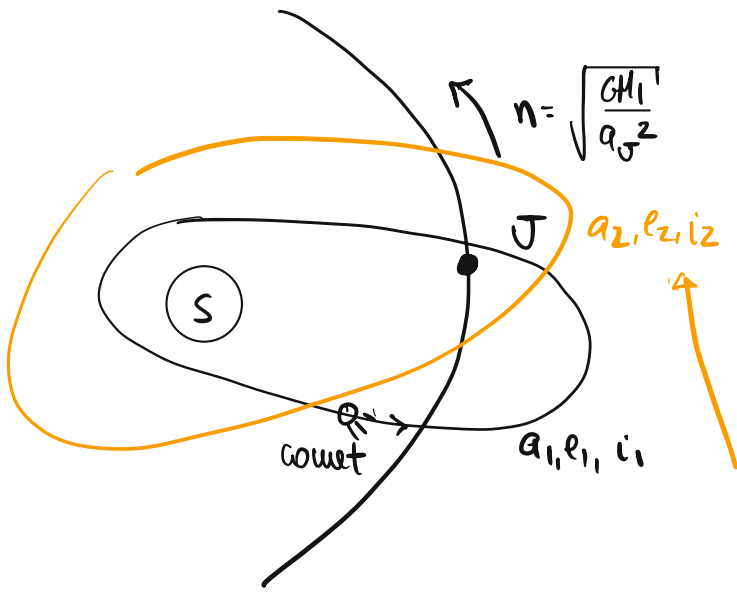


- If  $\mu \ll 1$  and if  $|\vec{r} - (1-\mu)\vec{R}| \gg 1$ , then we can say that  $U(\vec{r}) \sim -GM_1/|\vec{r} + \mu \vec{R}| + O(\mu)$

$$\Rightarrow J = \frac{1}{2} v^2 - \frac{GM_1}{|\vec{r}_B|} + O(\mu)$$

2-Body Energy

garbage close  
to Jupiter



$$J = - \frac{GM_1}{2a} - n \hat{z} \cdot \vec{h}$$

$a \equiv$  semi-major axis of comet orbit about the Sun

new comet orbit after Jupiter flyby

where

$$\vec{h} = h \hat{h} = \sqrt{GM_1 a (1 - e^2)} \hat{h}$$

$$\hat{z} \cdot \hat{h} = \cos i$$

$e \equiv$  eccentricity of the comet

$i \equiv$  inclination of the comet

- Writing all of the pieces:

$$J = - \frac{GM_S}{2a_1} - n \sqrt{GM_S a_1 (1 - e_1^2)} \cos i_1 = - \frac{GM_S}{2a_2} - n \sqrt{GM_S a_2 (1 - e_2^2)} \cos i_2$$

Tisserand Criterion

Tisserand Parameter

Links orbit elements  $(a, e, i)$  to the central body before and after an interaction with the secondary body

- Energy difference of the flyby

$$J = E_1 - n \vec{h}_1 \cdot \hat{z} = E_2 - n \vec{h}_2 \cdot \hat{z}$$

$$\Rightarrow \Delta E = n \Delta \vec{h} \cdot \hat{z}$$

$$\Delta E = E_2 - E_1 = n (\vec{h}_2 - \vec{h}_1) \cdot \hat{z} = n \Delta \vec{h} \cdot \hat{z}$$

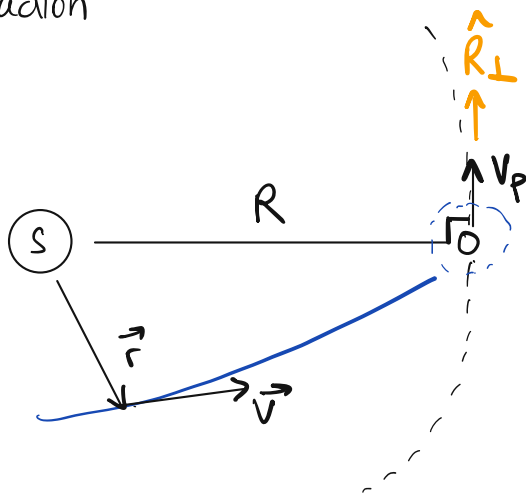
of the flyby

Q: When does a planet's gravitational attraction become negligible / important relative to the central body?

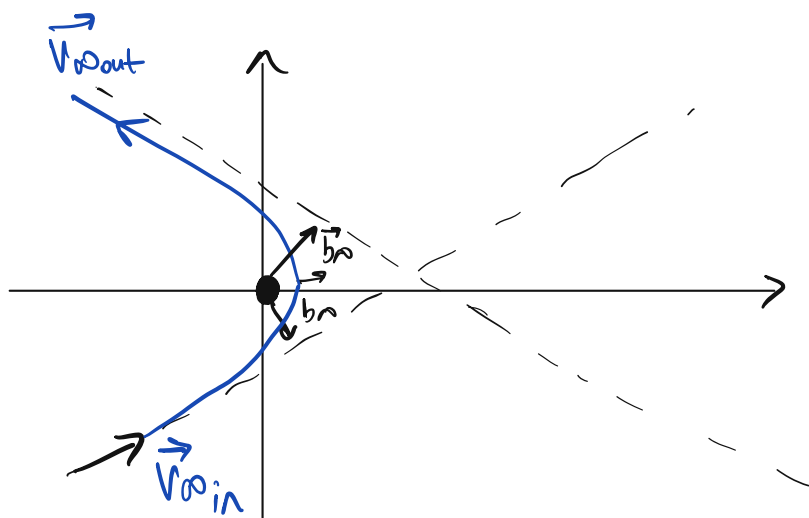
A: Sphere of influence  $\Rightarrow$  defines the distance and orbit parameters when we need to consider both bodies or just one of the bodies.  
Page 198 Roy a partial discussion of it.

### IV. GRAVITY ASSIST

- Can only occur in a restricted 3BP.
- We are only looking at "zeroth order" gravity assist.
- 2 phases of interaction

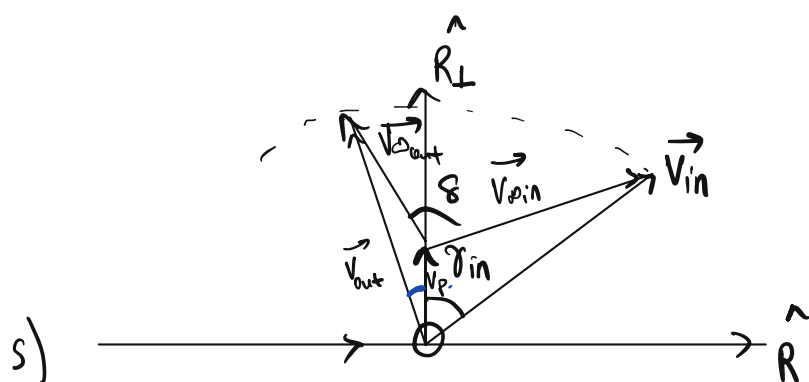


- Assume S/C travels outside of the planet's SOI
- Zeroth order approx  $\Rightarrow$  consider the planet's SOI to be "0", until we get "on top of it", then we switch to a SOI to be "∞" (No sun considered)
- Design S/C trajectory such that  $\vec{r} = \vec{R} + \vec{b}_\infty$  ;  $|\vec{b}_\infty| \ll R$ ,

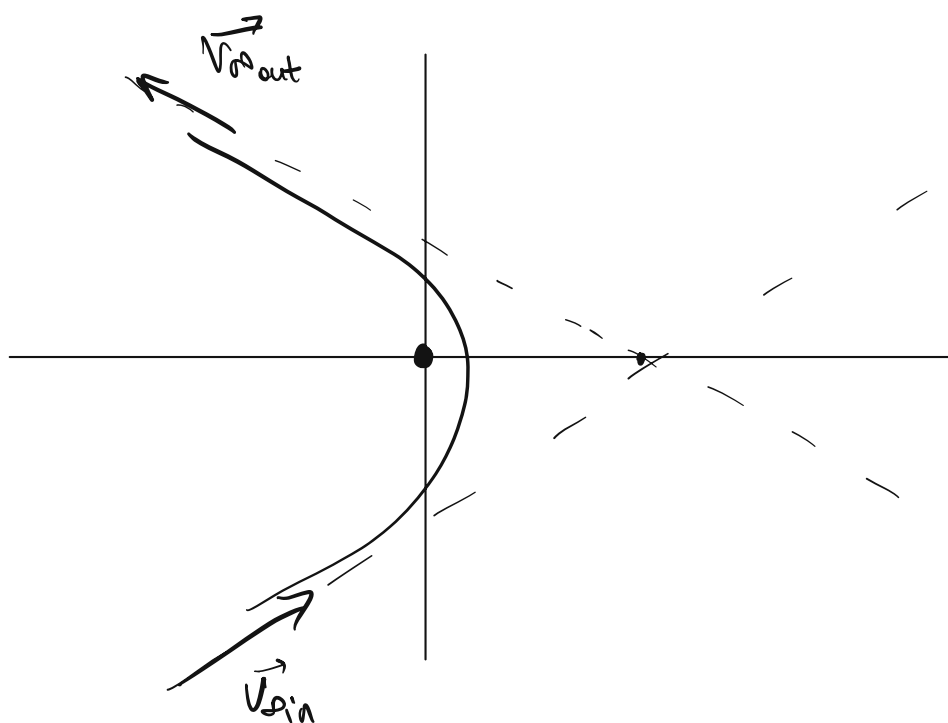


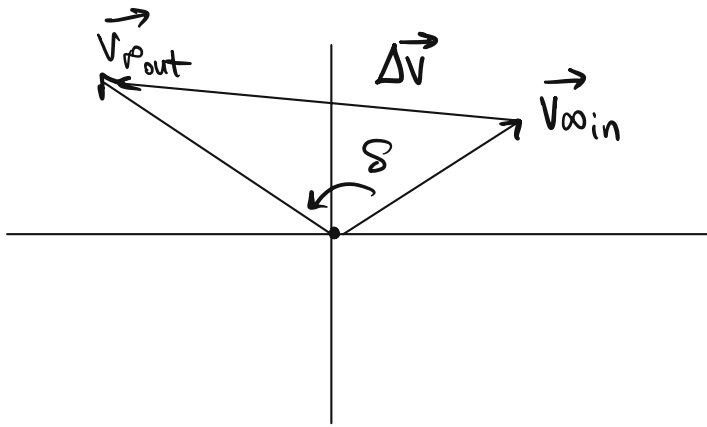
$$\vec{v}_{\infty in} = \vec{v} - v_p \hat{R}_\perp$$

$$\vec{b}_{\infty} = \vec{r} - \vec{R}$$

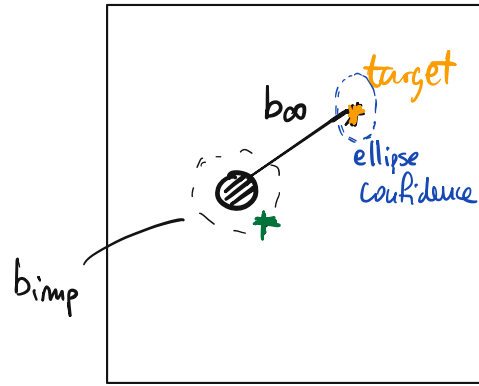


$$\vec{r}_{in} \sim \vec{R} \sim \vec{r}_{out}$$

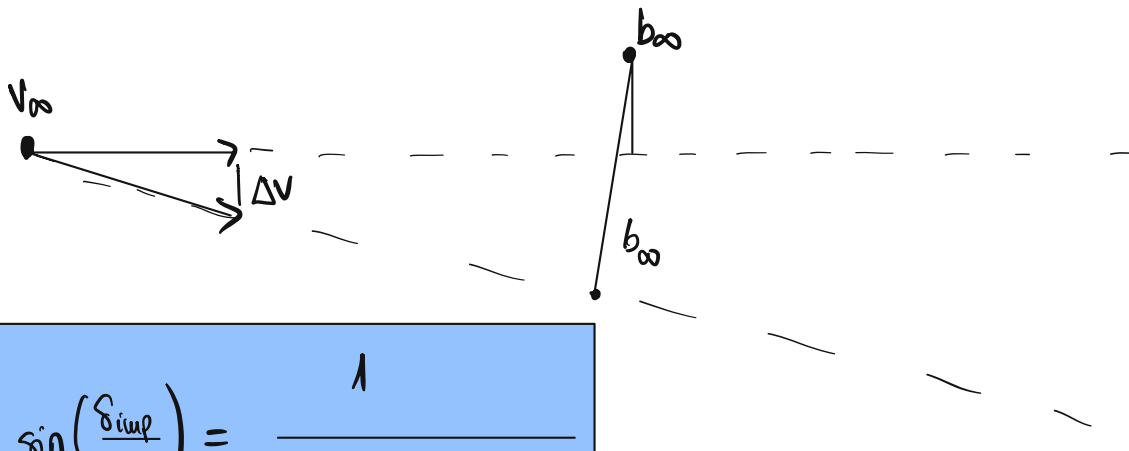




B-plane



$$\sin\left(\frac{\delta}{2}\right) = \frac{1}{\sqrt{1 + \frac{V_{\infty}^4 b_{\infty}^2}{(GM_p)^2}}}$$

Function of  $V_{\infty}, M_p, b_{\infty}$ Can only easily modify  $b_{\infty}$ 

$$\sin\left(\frac{\delta_{\text{imp}}}{2}\right) = \frac{1}{1 + \frac{V_{\infty}^2 R_p}{GM_p}}$$

maximum turn angle  $\delta_{\text{max}}$ 

\* Entry and Landing on Planet.

## V. EQUILIBRIUM POINTS

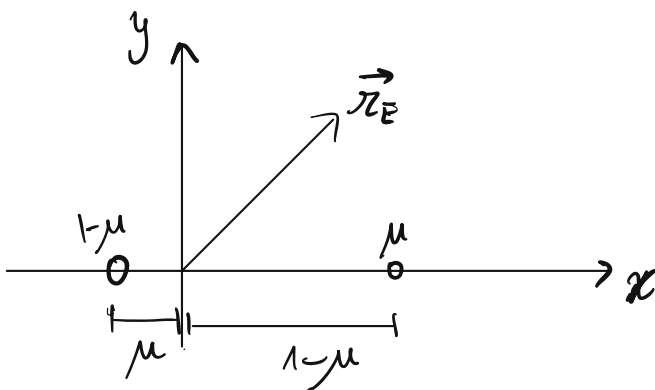
- Non-dimensional form

$$x'' - 2y' = \partial V / \partial x$$

$$y'' + 2x' = \partial V / \partial y$$

$$z'' = \partial V / \partial z$$

$$V = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{\sqrt{(x+\mu)^2 + y^2 + z^2}} + \frac{\mu}{\sqrt{(x-1+\mu)^2 + y^2 + z^2}}$$



- Equilibrium points in the rotating frame are of practical interest.

- A stationary point in the rotating frame.

- Defined by  $x' = y' = x'' = y'' = 0$  for a given location  $(x_E, y_E, z_E)$

- Requirement is that  $V_x = V_y = V_z = 0$  at  $\vec{r}_E$

$$V_x = x - \frac{1-\mu}{r_\mu^3} (x+\mu) - \frac{\mu}{r_{1-\mu}^3} (x-1+\mu)$$

$$V_y = y - \frac{(1-\mu)}{r_\mu^3} y - \frac{\mu}{r_{1-\mu}^3} y$$

$$V_z = -\frac{(1-\mu)}{r_\mu^3} z - \frac{\mu}{r_{1-\mu}^3} z$$

$$r_\mu = \sqrt{(x+\mu)^2 + y^2 + z^2} = r_{13}$$

$$r_{1-\mu} = \sqrt{(x-1+\mu)^2 + y^2 + z^2} = r_{23}$$

$$r_\mu, r_{1-\mu} > 0$$

$$1-\mu > 0, \mu > 0$$

- $V_z = -\left[ \underbrace{\frac{(1-\mu)}{r_\mu^3} + \frac{\mu}{r_{1-\mu}^3}}_{>0} \right] z$ ; when is  $V_z = 0$ ?  $\Rightarrow z = 0$

$\Rightarrow$  All equilibria are in x-y plane



$$\bullet V_y = \left[ 1 - \frac{(1-\mu)}{r_\mu^3} - \frac{\mu}{r_{1-\mu}^3} \right] y = 0$$

Either  $y=0$  or

$$1 - \frac{(1-\mu)}{r_\mu^3} - \frac{\mu}{r_{1-\mu}^3} = 0$$

$$\bullet V_x = x - \frac{(1-\mu)}{r_\mu^3} (x+\mu) - \frac{\mu}{r_{1-\mu}^3} (x-1+\mu)$$

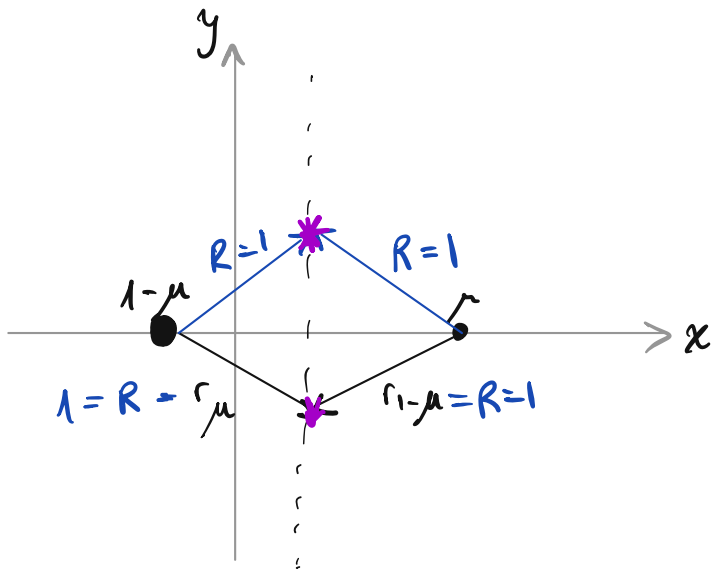
$$= x \left[ 1 - \frac{(1-\mu)}{r_\mu^3} - \frac{\mu}{r_{1-\mu}^3} \right] - \frac{\mu(1-\mu)}{r_\mu^3} + \frac{\mu(1-\mu)}{r_{1-\mu}^3} = 0$$

$$(A) \quad 1 - \frac{(1-\mu)}{r_\mu^3} - \frac{\mu}{r_{1-\mu}^3} = 0$$

$$(B) \quad y = 0$$

Take (A) first:

$$\mu(1-\mu) \left[ \frac{1}{r_{1-\mu}^3} - \frac{1}{r_\mu^3} \right] = 0 \Rightarrow r_\mu = r_{1-\mu}$$



Condition (A)

$$1 - \frac{(1-\mu)}{r_\mu^3} - \frac{\mu}{r_{1-\mu}^3} = 0$$

$$r_\mu = r_{1-\mu} = R \Rightarrow 1 - \frac{1}{R^3} (1-\mu + \mu) = 0$$

$$\Rightarrow R = 1$$

Lagrange Equilateral Triangle

Equilibrium points

$$z = 0 \quad r_\mu^2 = r_{1-\mu}^2 = 1$$

- Expanding radius:

$$\ominus \quad \begin{aligned} x^2 + 2\mu x + \mu^2 + y^2 &= 1 \\ x^2 - 2(1-\mu)x + (1-\mu)^2 + y^2 &= 1 \end{aligned}$$

$$2\mu x + 2(1-\mu)x + \mu^2 - (1-\mu)^2 = 0 \Rightarrow x_{eq} = \frac{1}{2}(1-2\mu) = \frac{1}{2} - \mu$$

$$\Rightarrow y^2 = 1 - (x+\mu)^2 = 1 - \left(\frac{1}{2} - \mu + \mu\right)^2 = \frac{3}{4} \Rightarrow y_{eq} = \pm \sqrt{3}/2$$

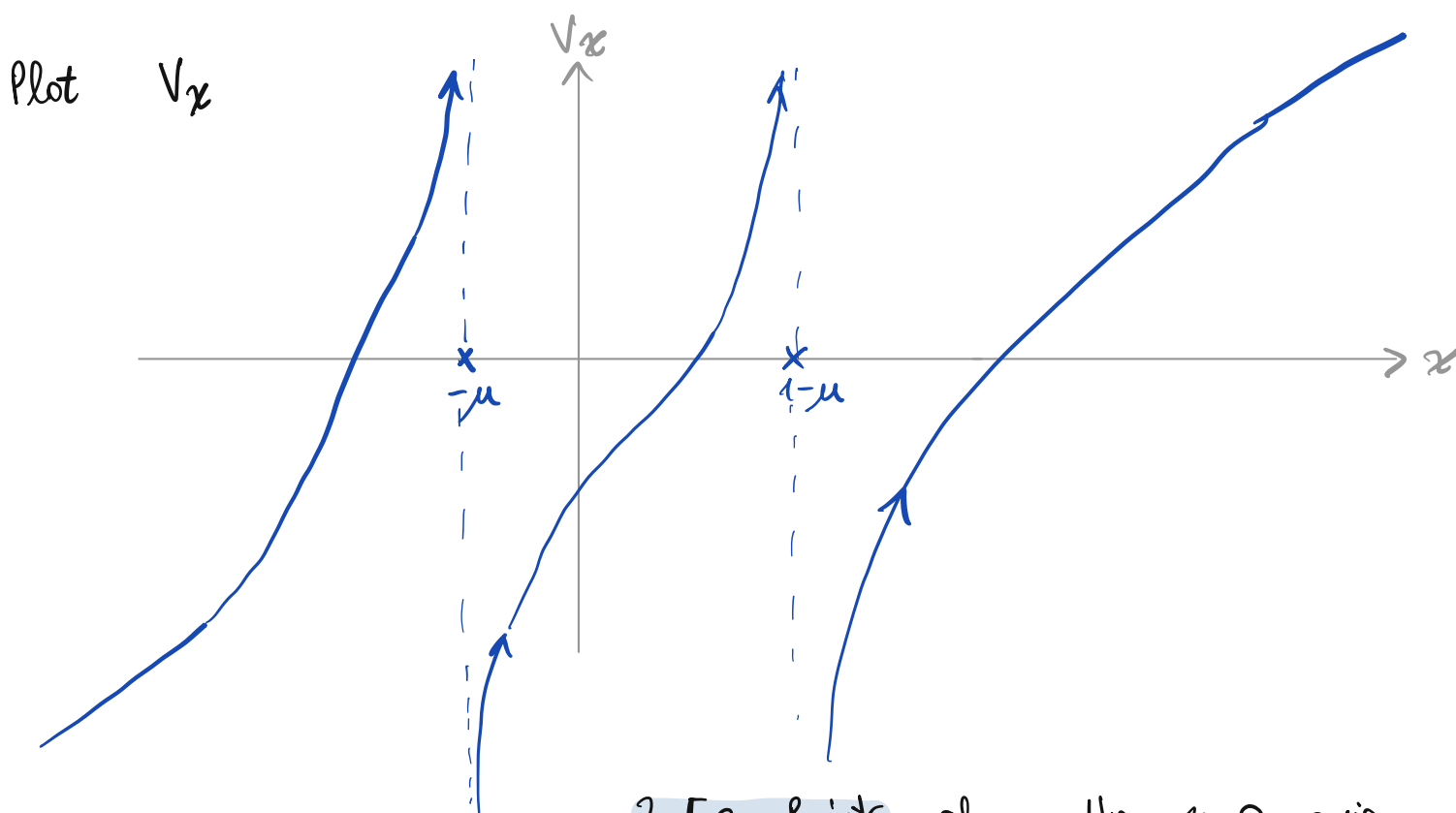
- Option (B)  $z=y=0$

$$V_x = x - \frac{(1-\mu)(x+\mu)}{r_\mu^3} - \frac{\mu(x-(1-\mu))}{r_{1-\mu}^3} = 0$$

$$r_\mu = \sqrt{(x+\mu)^2} = |x+\mu|$$

$$r_{1-\mu} = \sqrt{(x-1+\mu)^2} = |x-1+\mu|$$

$$\Rightarrow V_x = x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} - \frac{\mu(x-(1-\mu))}{|x-(1-\mu)|^3} = 0$$



3 Eq. points along the  $y=0$  axis

|          |       |       |                  |                                                                     |
|----------|-------|-------|------------------|---------------------------------------------------------------------|
| Roy      | $L_1$ | $L_2$ | $L_3$            | } Don't rely on notation, ask geometrically what convention is used |
| Szebekey | $L_3$ | $L_2$ | $L_1$            |                                                                     |
| Pollard  | $L_1$ | $L_3$ | $L_2$            |                                                                     |
| Darby    | $L_3$ | $L_1$ | $L_2$ ← Standard |                                                                     |

- Locations of  $L_{1,2,3}$ : Expand in orders of  $\mu^{1/3}$

$$\begin{aligned} L_1: x_1 &\sim 1 - (1/3)^{1/3} \mu^{1/3} + \mathcal{O}(\mu^{2/3}) \\ L_2: x_2 &\sim 1 + (1/3)^{1/3} \mu^{1/3} + \mathcal{O}(\mu^{2/3}) \\ L_3: x_3 &\sim -1 + \mathcal{O}(\mu) \end{aligned}$$

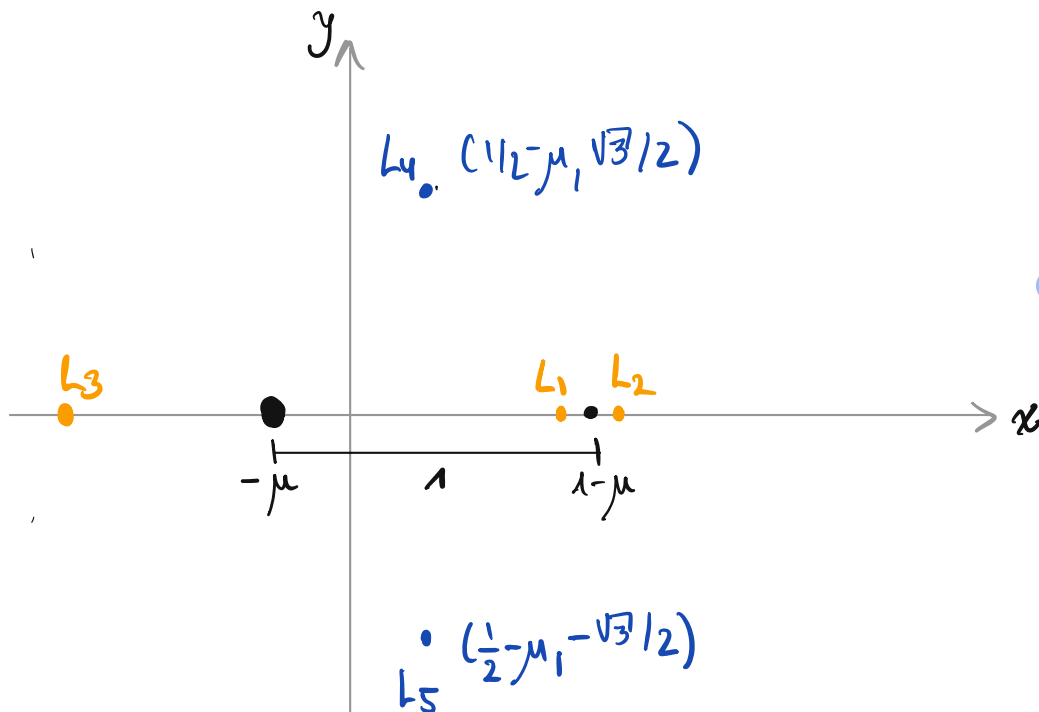
$$(1/3)^{1/3} \sim 0.69...$$

only valid if  $\mu \ll 1$

$$\mu_{EM} \sim 0.01, \mu_{EM}^{1/3} \sim 0.215... \text{ NO}$$

$$\mu_{ES} \sim 10^{-6}, \mu_{ES}^{1/3} \sim 0.01 \text{ YES}$$

$$L_{4,5}: x_{4,5} = \frac{1}{2} - \mu, y_{4,5} = \pm \sqrt{3}/2$$

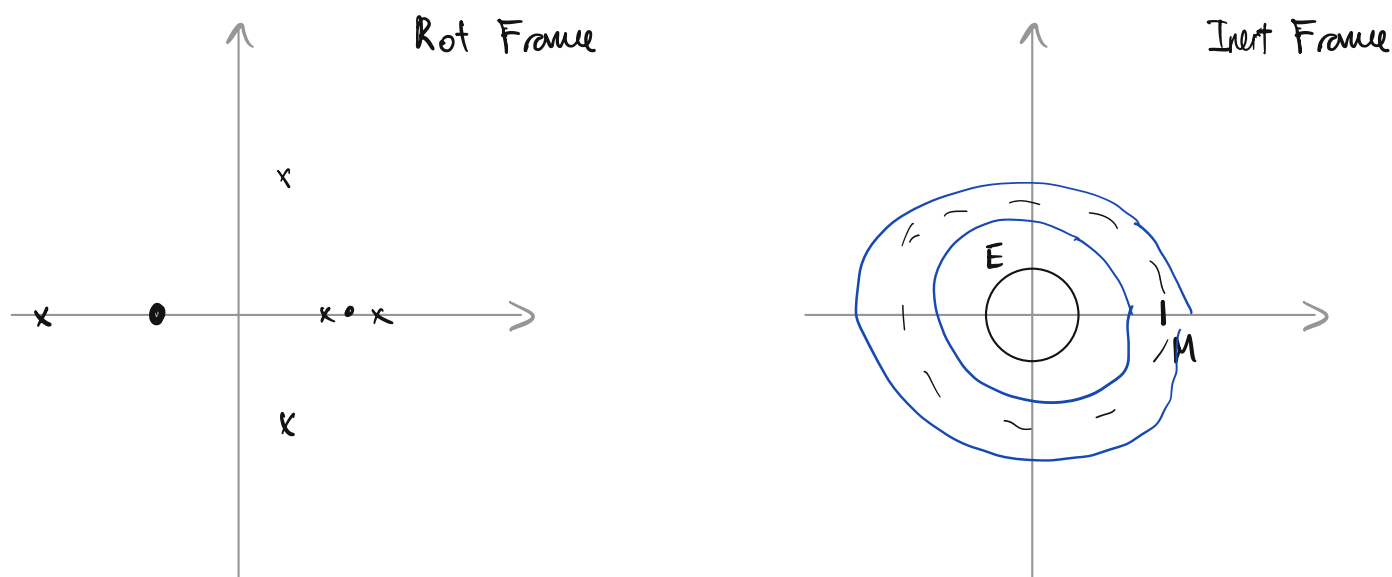


$L_i \equiv$  Libration points

$L_{4,5} \equiv$  Lagrange pts

$L_{1,2,3} \equiv$  Euler eq pts

- $L_{1,2,3,4,5}$  pts are really circular orbits in inertial space



## VI. ZERO VELOCITY SURFACES

- Let's take a different look at Eq pts + Jacobi integral

$$J = \frac{1}{2} v^2 - V(x, y, z) \quad V(x, y, z) = \frac{1}{2} (x^2 + y^2) + \frac{1-\mu}{r_\mu} + \frac{\mu}{r_{1-\mu}}$$

- Jacobi integral places constraints on where a particle can go:

$$\frac{1}{2} v^2 = J - V(x, y, z) \geq 0 \Rightarrow \text{when } J + V = 0$$

Recall ZBP  $\frac{1}{2} v^2 = E + \frac{\mu}{r} \geq 0$

$\Rightarrow$  defines Zero Velocity Surface

$\Rightarrow$  Motion only possible on one side of the surface, which is where  $v^2 > 0$

- If  $v^2 < 0 \Rightarrow$  Imaginary speed!  $\Rightarrow$  Quantum

- Note that  $V(x, y, z) > 0 \quad \forall (x, y, z)$ , never  $= 0$  except when  $x=y=0, z \rightarrow \pm \infty$

## V.I. General Intuition

- What constraints exist as a function of  $J$ . Let  $J = -C$

(i) If  $J \geq 0, C \leq 0$ , then there are no constraints

$$\frac{1}{2} v^2 = J + V > 0$$

(ii) If  $J < 0, C > 0$

$$\frac{1}{2} v^2 = -C + V$$

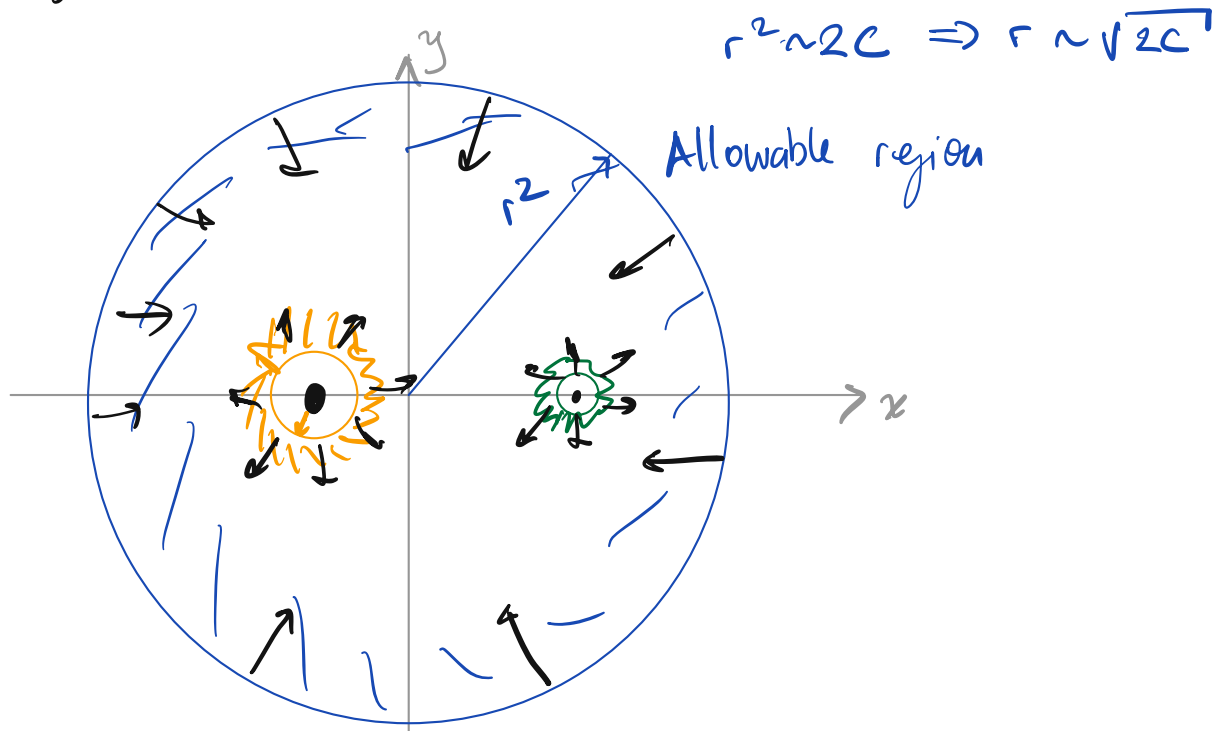
If we solve for  $V(x, y, z) = C > 0 \Rightarrow$  gives Zero Velocity Curves

Consider  $C \gg 1$

$$C = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_\mu} + \frac{\mu}{r_{1-\mu}} \gg 0$$

3 ways for  $V$  to be large

(1)  $x^2 + y^2 \sim r^2 \gg 1$   $\mu$



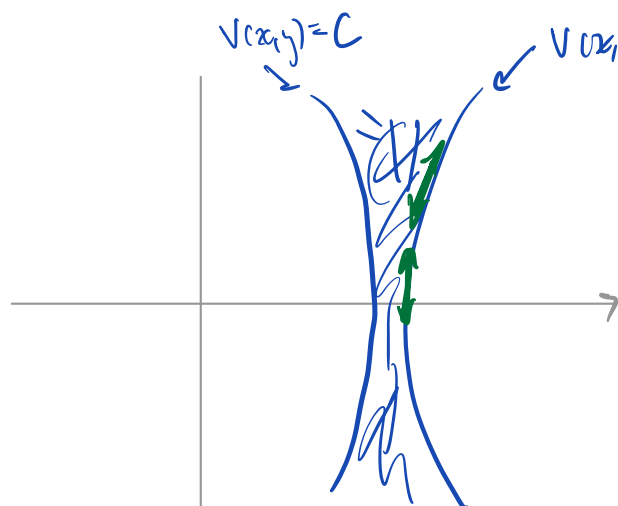
(2)  $r_\mu \ll 1$

$$r_\mu \sim \varepsilon \Rightarrow \frac{1-\mu}{\varepsilon} \sim C \Rightarrow \varepsilon \sim \frac{1-\mu}{C} \ll 1 \quad \text{?}$$

(3)  $r_{1-\mu} \ll 1$

$$\frac{\mu}{\varepsilon} \sim C \Rightarrow \varepsilon \sim \mu/C \ll 1$$

- For a large  $C$ , we get 3 realms of motion, all separated from each other.
  - As  $C \downarrow$ , outer curve start to shrink and inner curves increase  $\frac{z}{2}$
  - As  $C \downarrow$ , ZVC come closer, and eventually will intersect
- When 2 ZVCS intersect,  $V_x = V_y = V_z = 0$  !



$$V(x_0 + \Delta x, y_0 + \Delta y) = C$$

$$\cancel{V(x_0, y_0)} + V_x \Delta x + V_y \Delta y + \dots = \cancel{C}$$

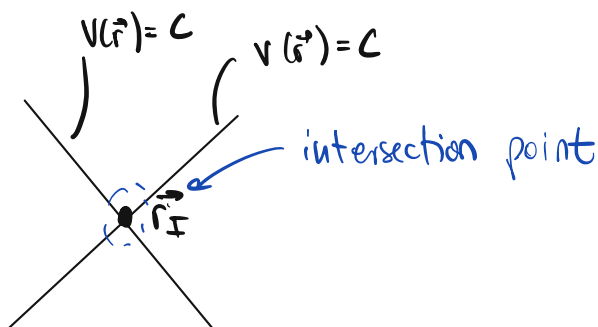
$$\Rightarrow \frac{\Delta x}{\Delta y} = -\frac{V_y}{V_x}$$



Intersection  $\Rightarrow$  4 solutions

$$\begin{aligned} V(x_0 + \Delta x, y_0 + \Delta y) = & \\ V|_0 + V_x \Delta x + V_y \Delta y & \\ + \frac{1}{2} V_{xx} \Delta x^2 + \frac{1}{2} V_{yy} \Delta y^2 + V_{xy} \Delta x \Delta y & \end{aligned}$$

not unique linear solution  $\Rightarrow V_x \Delta x + V_y \Delta y = 0$



$$V(\vec{r}_I) = V(x_I, y_I)$$

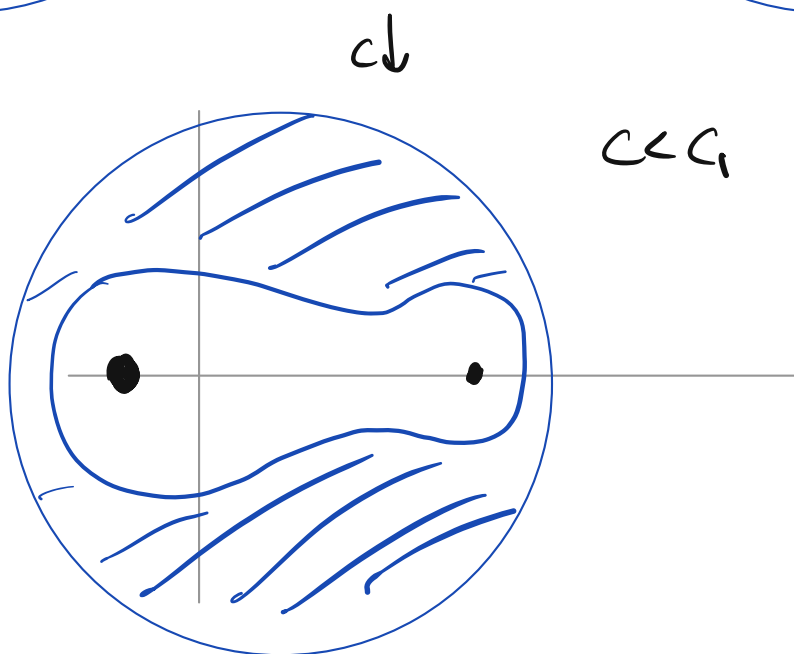
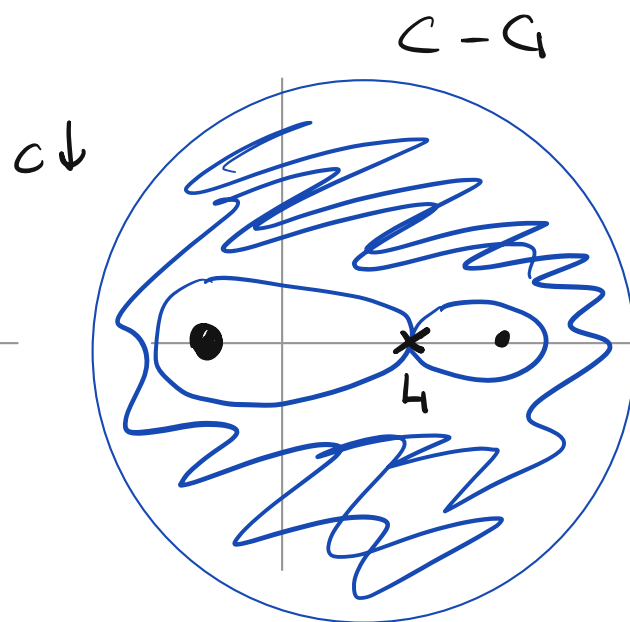
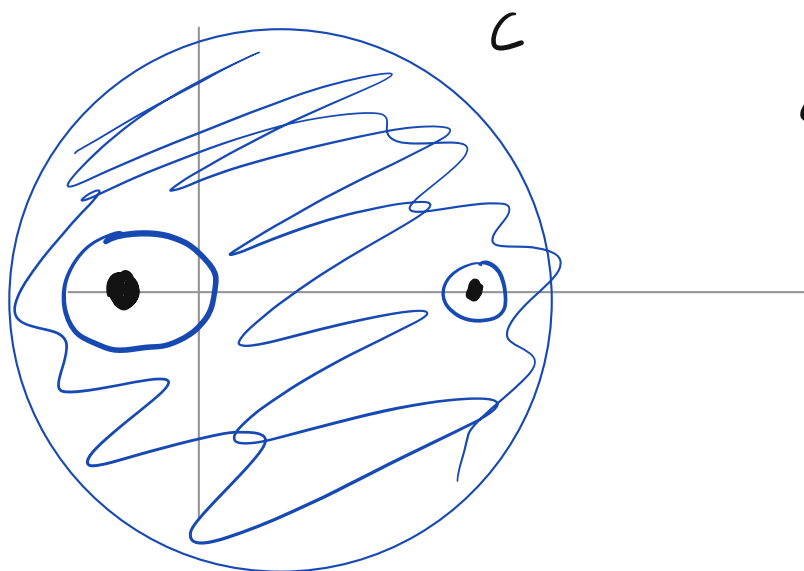
$$\cancel{C} = V(x_I + \delta x, y_I + \delta y) = \cancel{V(x_I, y_I)} + \frac{\partial V}{\partial x} \Big|_I \delta x + \frac{\partial V}{\partial y} \Big|_I \delta y$$

not unique (topologically)

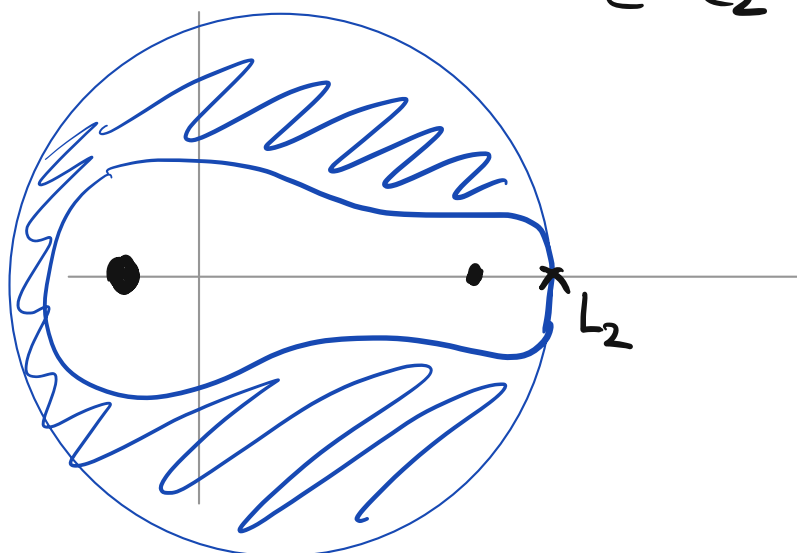
$$+ \frac{1}{2} [V_{xx}|_I \delta x^2 + 2V_{xy}|_I \delta x \delta y + V_{yy}|_I \delta y^2]$$

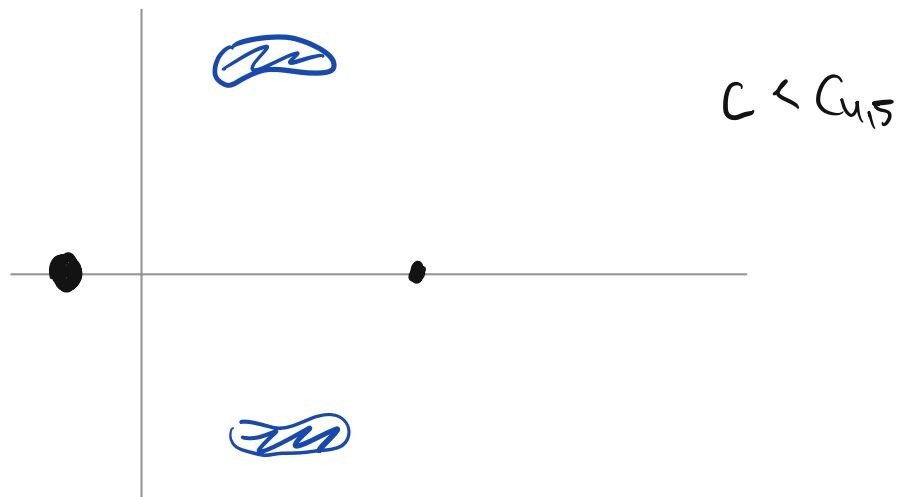
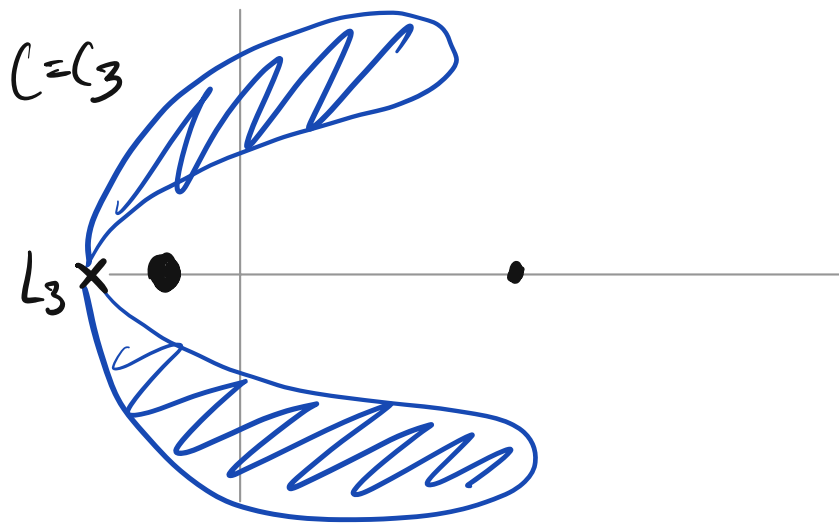
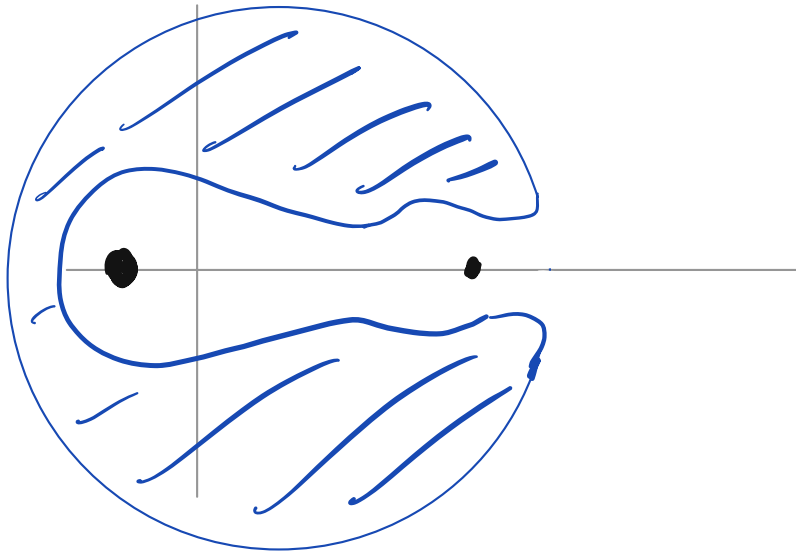
- Point is, when the ZVS intersect, the intersection point is at the Equilibrium points of the system.

# II. ZVS for different energies



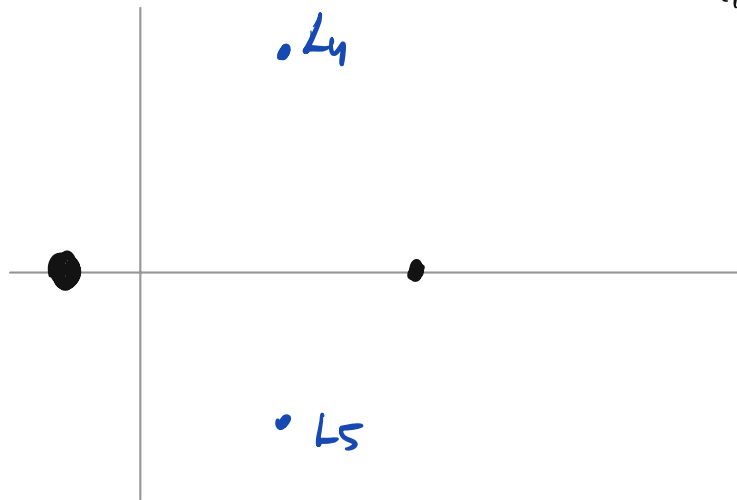
$C < C_1$







$$C = C_{4,5}$$



- The Jacobi Energy of the Eq. Pts play an important role in controlling what is possible in terms of orbit transfers between regions.

$$J = \frac{1}{2}v^2 - V(x,y,z) ; \quad J_i = -V(x_i, y_i, z_i) = -C_i \quad (0 \text{ velocity})$$

| $J$                   | Restrictions                                                     |
|-----------------------|------------------------------------------------------------------|
| $J < -C_1$            | Orbits about $P_1, P_2$ and far from system can't switch regions |
| $-C_1 < J < -C_2$     | Transfers between $P_1$ and $P_2$ can occur, but not to exterior |
| $-C_2 < J < -C_3$     | We can transfer between all regions, with restrictions           |
| $-C_3 < J < -C_{4,5}$ | We cannot get close to $L_4, L_5$ positions                      |
| $-C_{4,5} < J$        | Can move anywhere on x-y plane, restrictions in the z directions |
| $J \geq 0$            | No restrictions on motion over all x, y, z space                 |